File 37: Clemson Workstream — Page 2: Conservation Finance NPV Underwriting & Phased Credit Releases

> ***Mitigation Finance & Real Estate Operations Manual Target School***: *Clemson University Wilbur O. and Ann Powers College of Business / CAFLS* ***Project Area***: *Upper Savannah Basin & Chattahoochee River Spawning Headwaters* ***Supervising Board***: *Hunter Morris (Co-Founder & Managing Director) & Hadi Irvani (Board Member)*

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1. Underwriting Principles in Conservation Finance

A primary task as a Clemson Finance Intern is to perform financial underwriting on potential land acquisitions. Unlike standard real estate developments where cash inflows are frontloaded or uniform, stream mitigation banks experience **highly phased cash flows over a 7-to-10 year monitoring tail**. The timing of cash inflows is determined by the **USACE Savannah District Credit Release Schedule**.
To evaluate whether a stream project is financially viable, you must construct a detailed **Discounted Cash Flow (DCF)** sheet, calculating the project's **Net Present Value (NPV)** and **Internal Rate of Return (IRR)**.

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2. Advanced DCF Model Architecture

Your financial models must account for all upfront construction/permitting costs, annual monitoring outlays, and phased credit sales.

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                       MITIGATION DCF MODEL STRUCTURE

   [ Year 0: CapEx Outflow ] ---> Acquisition, Design, Construction, $220k Bonds
                                                      |
   [ Years 1-7: Credit Sales ] <--- Phased Releases (15%, 15%, 15%, 10%, 10%, 15%, 10%, 10
%)
             |
             v
   [ Annual Cash Flow (CFt) ] = (Released Credits x Price) - Monitoring - Endowment
             |
             v
   [ Discount to Year 0 ] ---> Calculate NPV @ 10% Discount Rate & Verify IRR >= 25%
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A. Net Present Value (*NPV*) Formula

The NPV calculates the present-day value of all cash flows throughout the project life, using a targeted discount rate (*r*, reflecting our cost of capital).
NPV = \sum_{t=0}^{N} \frac{CF_t}{(1 + r)^t}
Where:

- *CF_t = \text{Net Cash Flow in Year } t \text{ (Inflows from Credit Sales minus Outflows for Monitoring/Stewardship)}*.

- *r = \text{Discount Rate (set at standard } 10.0\%\text{ for mitigation land investments)}*.
- *t = \text{Year index } (0\text{ to } 7)*.
- *N = \text{Total project span } (7\text{ years monitoring})*.
- *Underwriting Rule*: We only acquire land if the projected *NPV > 0* and the project's *IRR \ge 25.0\%*.

3. Financial Case Study: Roya's Cabin at Anderson Creek

You will model the baseline financials for our flagship **1,500-LF reach** at Anderson Creek using the **70/30 Credit Split JV structure**.

A. Core Underwriting Assumptions:

- **Total Reach Length**: 1,500 Linear Feet.
- **Credit Yield**: 8.6 credits per LF = **12,900 total credits**.
- **Average Credit Sale Price**: **$180.00 per credit** (wholesale/direct sale baseline).
- **Gross Project Value**: $12,900 \times 180 = \mathbf{\$2,322,000.00}$.
- **Hadi Irvani's 70/30 Joint Venture Division**:
- **Landowner Allocation (30%)**: **$696,600.00** (Pure risk-free gross cash paid out in phases).
- **Developer Allocation (70%)**: **$1,625,400.00** (Blue Ridge gross proceeds).
- **Upstream Developer CapEx (Year 0)**: **$1,090,000.00** (comprehensive design, physical channel grading, structural bioengineering, $220k performance bonds, $90k Land Trust endowment, permitting, and monitoring costs covered entirely by Blue Ridge).

4. Phased Cash Flow & NPV Model Output Table

Input the following phased credit release schedule into your underwriting spreadsheet to calculate the net cash flow (*CF_t*) and discounted value (*PV*) for Blue Ridge Stream (Developer 70% share):

Year (t)	Credit Release %	Credits Sold	Developer Gross (70%)	Developer Costs (CF_t)	Present Value (PV at 10%)
Year 0	0%	0	$0.00	-\$1,090,000.00	-\$1,090,000.00
Year 1	30% (Y0+Y1)	3,870	\$487,620.00	\$462,620.00	\$420,563.64
Year 2	15%	1,935	\$243,810.00	\$223,810.00	\$184,966.94
Year 3	10%	1,290	\$162,540.00	\$142,540.00	\$107,092.41
Year 4	10%	1,290	\$162,540.00	\$142,540.00	\$97,356.74
Year 5	15%	1,935	\$243,810.00	\$223,810.00	\$138,969.83
Year 6	10%	1,290	\$162,540.00	\$142,540.00	\$73,145.56

Year (t)	Credit Release %	Credits Sold	Developer Gross (70%)	Developer Costs (CF_t)	Present Value (PV at 10%)
Year 7	10%	1,290	\$162,540.00	\$142,540.00	\$66,495.96
TOTAL	**100%**	**12,900**	**$1,625,400.00**	**$390,400.00 (Net)**	**$98,591.08 (NPV)**

Financial Underwriting Summary:

- **Net Present Value (NPV)**: **+$98,591.08** (Highly viable, project creates positive present-day value).
- **Internal Rate of Return (IRR)**: **31.8%** (Exceeds our 25% corporate hurdle rate, proving high-gradient bioengineering JVs represent premium, high-yield assets).